

Ji Chengwang

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[Google scholar profile](#)



Education

University of Macau, Macau, China	2022.08 – 2026.07
- Ph.D. in Electrical and Computer Engineering - Supervisor: Prof. Ma Shaodan, Associate Director of State Key Laboratory of Internet of Things for Smart City	
Jilin University, Changchun, China	2018.08 – 2022.06
Bachelor in Communication Engineering	

Research Interests

- AI for Communication
- Integrated Sensing and Communication
- Physical Layer Communication
- Reconfigurable Antenna and Surface for 6G

Publications

AI for Communication

- Chengwang Ji, Haiquan Lu, Kehui Li, Qiaoyan Peng, Jintao Wang, and Shaodan Ma*, “Model-driven deep learning enhanced joint beamforming and mode switching for RDARS-aided MIMO systems,” *IEEE Trans. Cogn. Commun. Netw.*, under review, 2026. (SCI Q1, IF 7.0)
- Chengwang Ji, Haiquan Lu, Qiaoyan Peng, Jintao Wang, and Shaodan Ma*, “Mode Switching for RDARS-Aided ISAC Systems: From Optimization to Deep Unfolding,” preparing to submit to *IEEE Trans. Wireless Commun.* (SCI Q1, IF 10.7)

Physical Layer Communication

- Chengwang Ji, Qing Xue, Haiquan Lu*, Jintao Wang, Qiaoyan Peng, Shaodan Ma* and Wei Zhang, “Reconfigurable codebook-based beamforming for RDARS-aided mmWave MU-MIMO systems,” *IEEE Trans. Wireless Commun.*, vol. 25, pp. 8417-8433, 2026. (SCI Q1, IF 10.7)
- Chengwang Ji, Haiquan Lu, Qiaoyan Peng, Jintao Wang, and Shaodan Ma*, “Joint sparsity and beamforming design for RDARS-aided systems,” *IEEE Wireless Commun. Lett.*, vol. 15, pp. 2934-2938, 2026. (SCI Q1, IF 5.5)
- Chengwang Ji, Qiaoyan Peng, Jintao Wang, Ziqian Pei, and Shaodan Ma*, “Channel-aware mode switching enhanced RDARS-aided downlink mmWave MIMO systems,” in Proc. IEEE International Conf. on Commun. (ICC), 2025. (Flagship conference)
- Qing Xue, Chengwang Ji, Shaodan Ma*, Jiajia Guo, Yongjun Xu, Qianbin Chen, and Wei Zhang, “A survey of beam management for mmWave and THz communications towards 6G,” *IEEE Commun. Surveys Tuts.*, vol. 26, no. 3, pp. 1520–1559, 3rd Quart., 2024. (SCI Q1, IF 46.7, Highly Cited Paper)
- Jintao Wang, Pingping Zhang, Chengzhi Ma, Chengwang Ji, Zheng Shi, Guanghua Yang, and Shaodan Ma*, “A new paradigm towards reconfigurable environment: Reconfigurable distributed antennas and reflecting surface,” *Journal of Communication and Information Networks*, 2026. (Cover Article)
- Chengwang Ji, Haiquan Lu*, Qiaoyan Peng, Jintao Wang, and Shaodan Ma*, “Subarray-based RDARS for near-field UAV systems: Joint mode selection and position design,” under preparation.

Integrated Sensing and Communication

- Chengwang Ji, Haiquan Lu, Qiaoyan Peng, Jintao Wang, and Shaodan Ma*, “Joint waveform and mode switching design for RDARS-aided integrated sensing and communication systems,” in Proc. IEEE International Conf. on Commun. (ICC) Workshops, 2026. (Flagship conference)
- Chengwang Ji, Haiquan Lu, Qiaoyan Peng, Jintao Wang, Feifei Gao, and Shaodan Ma*, “Mode Switching for RDARS-Aided ISAC Systems: From Optimization to Deep Unfolding,” *IEEE Transactions on Communications*, under review, 2026.
- Jintao Wang, Chengwang Ji, Jiajia Guo, and Shaodan Ma*, “Demo: Reconfigurable distributed antennas and reflecting surface (RDARS)-aided integrated sensing and communication system,” in Proc. IEEE International Conf. on Commun. in China (ICCC), 2023, pp. 1–2.

Projects

Integrated Sensing and Communications for the Low Altitude Economy Development in the Guangdong-Hong Kong-Macao Greater Bay Area	2025.07 – 2028.06
Core member — FDCT-MOST project (0114/2025/AMJ)	
Responsibilities: Sensing scheme design and prototype platform development for multimodal monitoring	
Novel Active-Passive Hybrid Intelligent Surface-Assisted Sensing and Communications	2025.07 – 2028.06

Core member — FDCT, Funding Scheme for Scientific Research and Innovation project (0020/2025/RIB1)	
Responsibilities: Design of channel state information acquisition scheme for active-passive hybrid intelligent surface-assisted sensing and communication system	
Active-Passive Hybrid Intelligent Surface-Assisted Active Detection Sensing Prototype Platform	2023.08 – 2024.08
Co-Principal Investigator	
Responsibilities: Signal extraction, system platform development, and validation platform testing	
Active-Passive Intelligent Surface-Assisted Integrated Sensing and Communication Prototype Platform	2023.08 – 2024.08
Co-Principal Investigator—Presented at the 2023 International Conf. on Commun. in China (ICCC)	
Responsibilities: Design sensing and positioning algorithms; establish system platforms; conduct platform verification testing	
The First 6G Intelligent Wireless Communication Systems Competition	2023
Core member—Excellent Award (9th/1588)	
- Competition Topic: Design of Wireless AI for Adaptive and Online Updating Requirements in Scenarios with Small Sample Conditions - Responsibilities: Design small-sample channel expansion network	

Academic Experience

Journal Reviewer: TWC, TCOM, WCM, China Communications, TVT, WCL, CL

Conference Reviewer: ICC, VTC, WCNC, WNSP

Conference Technical Program Committee (TPC) Member

- 2026 IEEE International Conf. on Commun. (ICC), ISAC track
- Workshop on Reconfigurable Electromagnetic Technologies for 6G Multi-Functional Networks at PIMRC 2026.
- VTC2026-Spring, signal processing for wireless communications, cooperative communication

Teaching Assistant

- M.Sc. Course: Advanced Topics in Telecommunications, Advanced Topics in Applied Probability and Statistics, Modeling and Theoretical Analysis for Communication Systems s
- B.Sc. Course: Principles of Communication Systems

Awards

UM PhD Assistantship	2022–2026
The Guangdong-Hong Kong-Macao Greater Bay Area (University of Macau) Taihu Scholarship	2026
Excellent Paper of the 2026 Sichuan-Chongqing Science and Technology Conference	2026